

Figure S1

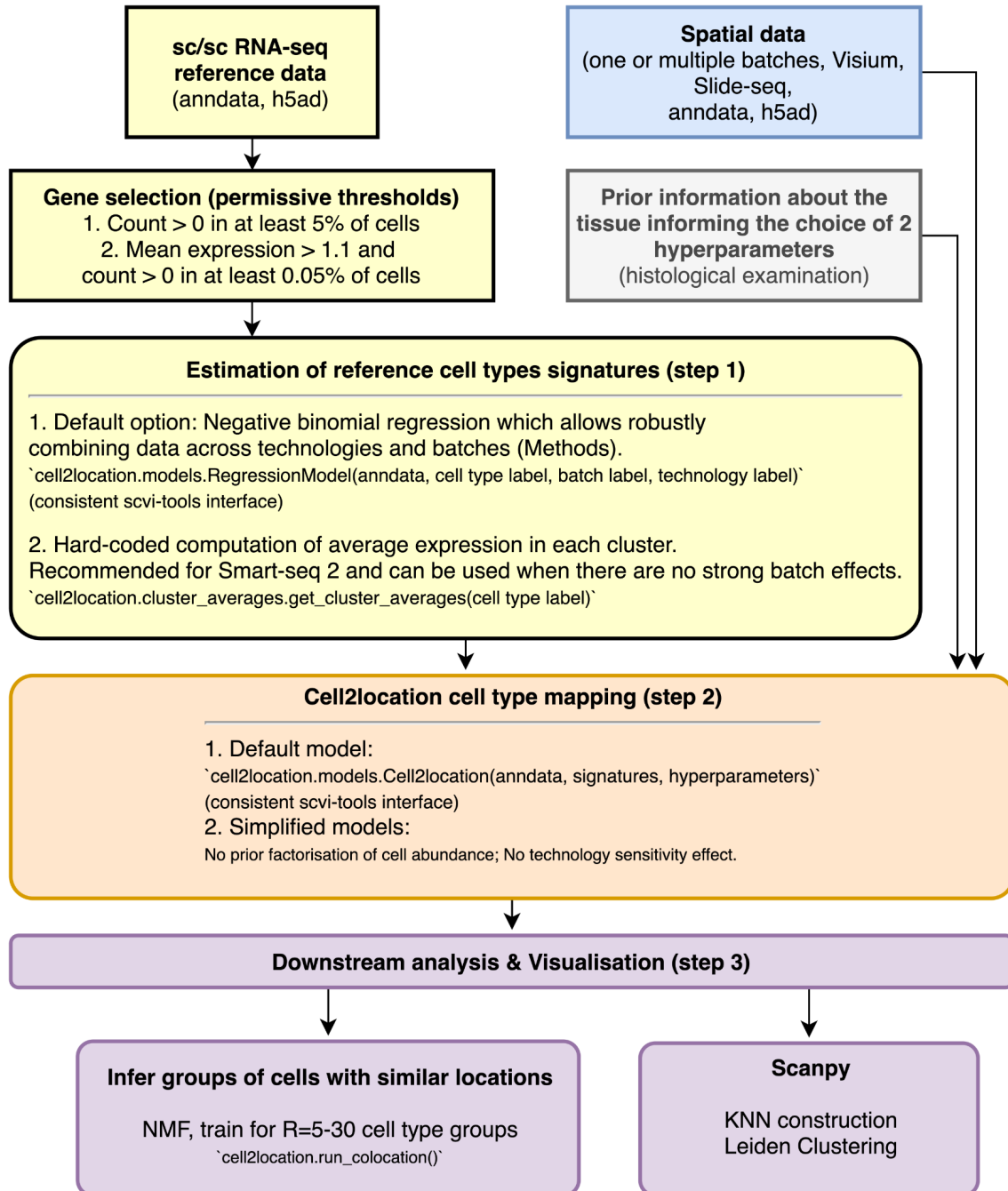


Figure S1. Details of cell2location workflow. Diagram of the cell2location workflow, which consists of estimation of reference cell type signatures estimation from scRNA-seq, cell2location cell type mapping and downstream analyses. Each block highlights relevant models, tutorial notebooks show the 3 steps outlined.

Figure S27

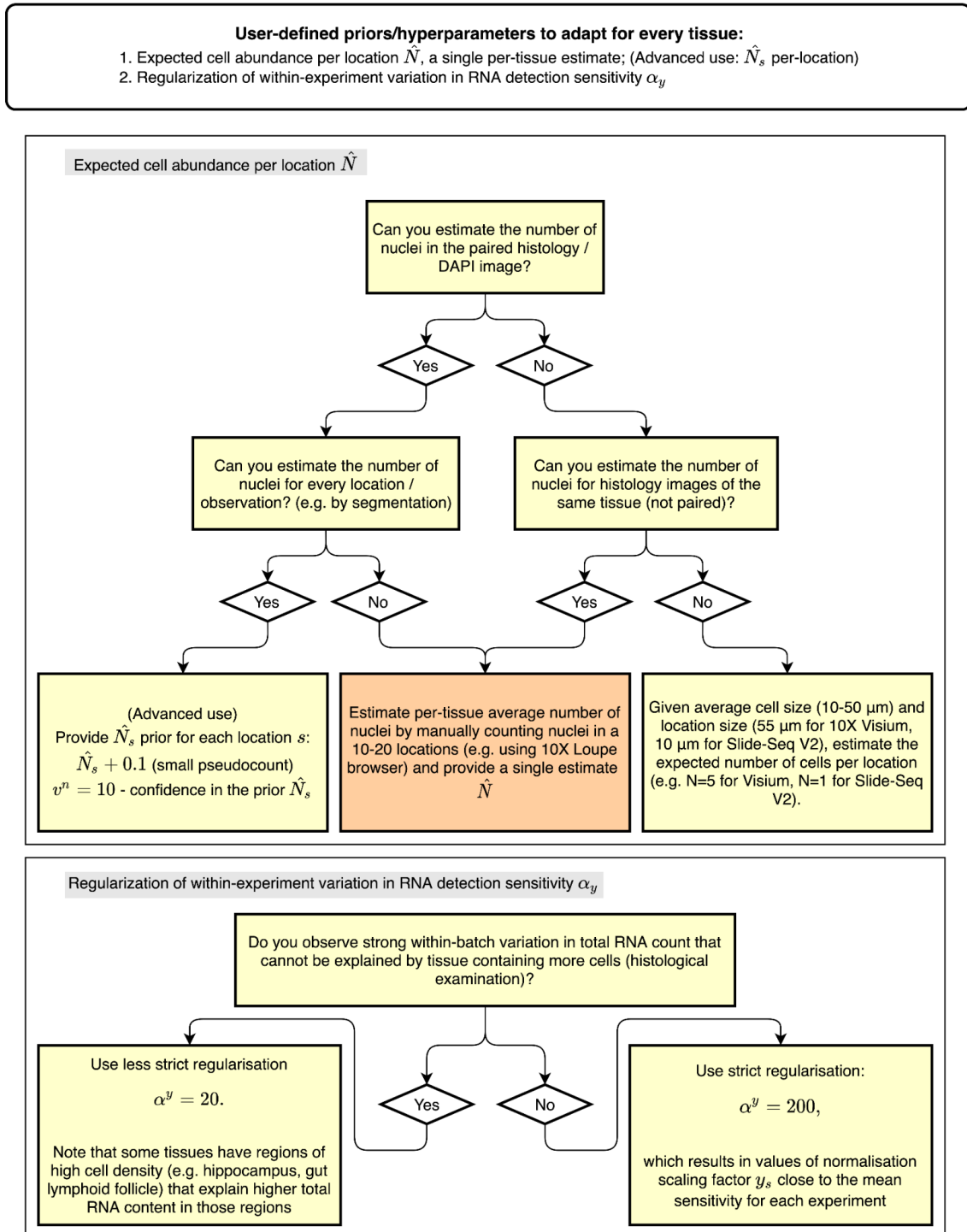


Figure S27. Flow diagram to choose cell2location hyperparameters. Cell2location has 2 hyperparameters that can be set by the users (see a separate box for each hyperparameter). The following flowchart can be used to determine these hyperparameters in the light of experimental or biological prior information. An appropriate setting of the expected cell abundance $\hat{N}$ is essential to inform the estimation of absolute cell type abundance values. Term ‘location’ refers to 10X Visium spots, Slide-Seq V2 beads, WTA Nanostring regions-of-

interest (ROI) and other spatially defined voxels where transcriptomic measurements are taken. The choice of expected cell abundance $\hat{n}$ highlighted in orange is applicable to most 10X Visium users.

Comprehensive mapping of tissue cell architecture via integrated single cell and spatial transcriptomics: cell2location model

Supplementary Methods

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Contents

1	Cell2location model	1
1.1	Model derivation	1
1.2	Determining hyper parameter values	7
1.3	Inference	8
1.4	Nuclei Segmentation (optional)	9
2	Estimation of reference expression signatures of cell types	9
3	Comparison of cell2location with alternative methods	11
4	Downstream analyses	12
4.1	Identification of tissue regions by clustering	13
4.2	Groups of co-located cell types	13
5	Constructing a synthetic spatial transcriptomics data set	14
6	Appendix	17
6.1	Complete graphical model diagram	17
6.2	Notation	17

1 Cell2location model

1.1 Model derivation

Cell2location is a Bayesian model that decomposes gene expression counts measured across spatial locations into a set of pre-defined reference cell types, thereby estimating both relative and absolute cell abundances. A representation of cell2location as graphical model is shown in Fig 1.

The input data of the method are as follows:

Let $D = \{d_{s,g}\}$ be a $S \times G$ count matrix with its entries corresponding to mRNA counts at locations $s = \{1, \dots, S\}$ from one or multiple batches, for genes $g = \{1, \dots, G\}$. The location index s iterates across unique locations across all batches $e = \{1, \dots, E\}$ (one location s belongs exclusively to one batch), where e denote individual experiments, such as 10X Visium chips (i.e. square capture areas) or Slide-Seq V2 pucks. Cell2location then jointly models the data across these spatial data sets $e = \{1, \dots, E\}$, aka batch integration, by accounting for technical effects aligned with batch (see below). When using cell2location combination with 10X Visium data, this matrix can be directly obtained from the 10X SpaceRanger software; see main text Methods.

Further, let $G = \{g_{f,g}\}$, denote an $F \times G$ matrix of reference cell type signatures, which consist of $f = \{1, \dots, F\}$ gene expression profiles $G_{f,:}$ for $g = \{1, \dots, G\}$ genes, which correspond to average expression levels of each gene in each cell type in a linear mRNA count scale. This matrix needs to be provided to cell2location and can be estimated from scRNA-seq profiles; see Section 2 for details.

We note that the matrices D and G need to be aligned such that they cover the same set of genes. The software implementation of cell2location builds on the Scanpy file formats, which provide native access to outputs of the 10X visium platform. For other technologies, such as Slide-Seq V2, files with count matrices of the appropriate formats need to be generated and imported into Scanpy data format by the user.

Cell2location models the elements of D as Negative Binomial distributed, given an unobserved expression level (rate) $\mu_{s,g}$ and a gene- and experiment- specific over-dispersion parameter $\alpha_{e,g}$, which accounts for residual variance in expression of individual genes that cannot be attributed to Poisson sampling noise:

$$d_{s,g} \sim \text{NB}(\mu_{s,g}, \alpha_{e,g}). \quad (1)$$

Equivalently, this negative binomial likelihood can be expressed as a Gamma-Poisson mixture with a Poisson likelihood (count measurement model) and a Gamma-distributed mean (expression dispersion model)¹. Parameterising the NB distribution by a Poisson rate parameter μ , and over-dispersion α , which corresponds to shape parameter of Gamma distribution, the equivalent Gamma-Poisson mixture follows as:

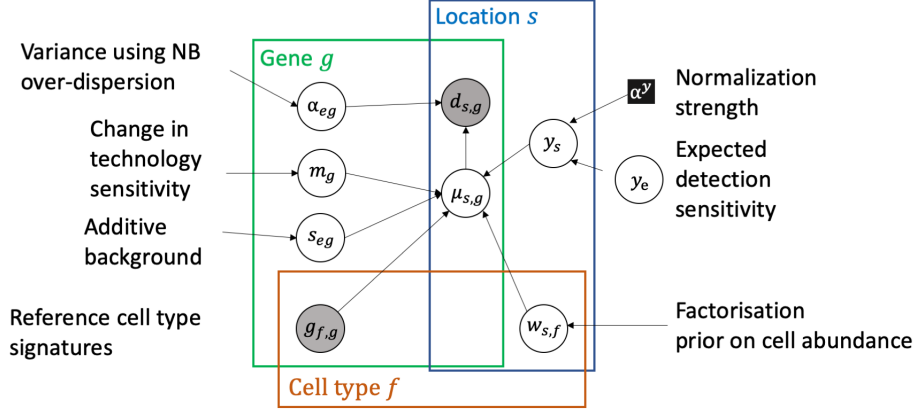
$$d_{s,g} \sim \text{Poisson}(\text{Gamma}(\alpha_{e,g}, \alpha_{e,g}/\mu_{s,g})). \quad (2)$$

The expression of level of genes $\mu_{s,g}$ in the mRNA count space is modelled as a linear function of expression signatures of reference cell types:

$$\mu_{s,g} = \left(\underbrace{m_g}_{\text{technology sensitivity}} \cdot \underbrace{\sum_f w_{s,f} g_{f,g}}_{\text{cell type contributions}} + \underbrace{s_{e,g}}_{\text{additive shift}} \right) \cdot \underbrace{y_s}_{\text{per-location sensitivity}}. \quad (3)$$

Here, $w_{s,f}$ denotes regression weight of each reference signature f at location s , which can be regarded as the abundance of cells expressing reference cell-type signature f at location s (denoted absolute cell abundance in the paper); m_g denotes a gene-specific scaling parameter, which adjusts for global differences in expression estimates between technologies (platform effect); $s_{e,g}$ captures gene-specific additive shift, e.g. due to free-floating RNA; and y_s denotes a location-specific scaling parameter, which models variation in RNA detection sensitivity across locations and experiments. The last parameter scales both the RNA contributions of cell types

A. mRNA counts model



B. Cell abundance prior

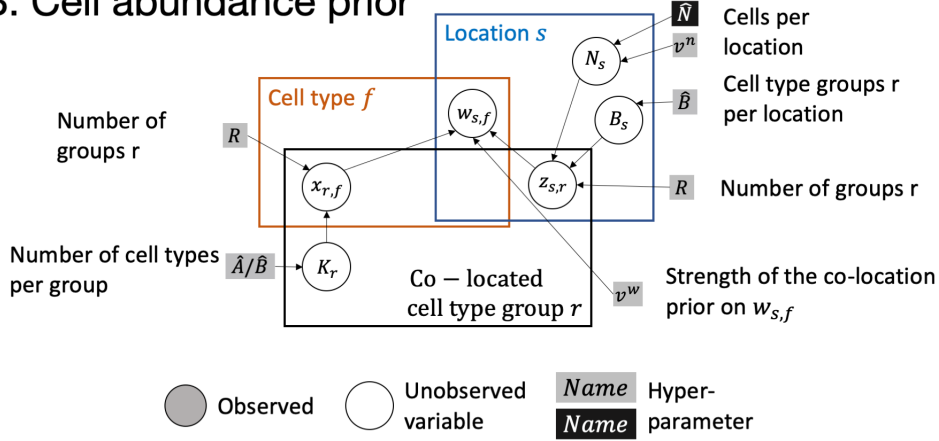


Figure 1: Graphical model illustration of the cell2location model. **A.** Main part of the model describing spatial mRNA counts. **B.** Additional layer of factorisation of $w_{s,f}$ representing similarity in cell type abundance across locations. Square boxes denote hyper-parameters, with black box highlighting the parameters that require adaptation to a specific data set (determined by tissue type and spatial technology), as mentioned in Section 1.2.

72 and contaminating RNA $s_{e,g}$.

73

74 The prior distributions on the unobserved parameters are as follows:

75 1. **Absolute abundance of cell types across locations.** The regression weights $w_{s,f}$ are
76 modelled as Gamma distributed

$$77 \quad w_{s,f} \sim \text{Gamma}(\mu_{s,f}^w v^w, v^w). \quad (4)$$

78 Here, v^w denotes a fixed hyper parameter that determines the prior strength. The prior
79 mean parameter $\mu_{s,f}^w$ is modelled in a hierarchical fashion, decomposing the regression
80 weights into R latent groups of cell types $r = \{1, \dots, R\}$, thereby accounting for linear
81 dependencies in spatial abundance of cell types:

$$82 \quad \mu_{s,f}^w = \sum_r z_{s,r} x_{r,f}. \quad (5)$$

83 The number of latent groups R is specified by the user (default $R = 50$). Intuitively, these
84 R groups of cell types can be thought of as cellular compartments (or zones) in the tissue,
85 which are characterized by shared cell type composition. We observed that the sensitivity
86 of mapping cell types with fine transcriptional differences in particular increases when
87 accounting for these dependencies (see Fig 5, Fig 4, Fig S25E, Fig S24B).

88 The prior distributions of $z_{s,r}$ and $x_{r,f}$ are defined to control absolute scale of the cell type
89 abundance estimates, guiding $w_{s,f}$ to the scale of the number of the abundance of cells
90 expressing reference cell type signature f at location s . This is necessary, because while
91 the scaling parameter m_g facilitates the integration across technologies and y_s corrects
92 detection sensitivity, it leads to non-identifiability between m_g , y_s and $w_{s,f}$, unless the
93 informative priors on all three variables are used. In addition, these prior distributions help
94 control the sparsity of how many cell types f are expected at each location s , facilitating
95 application of cell2location to tissues and technologies with varying numbers of cells and
96 cells types per location. The hyper-parameters controlling the $w_{s,f}$ prior can be estimated
97 from paired histology image, if available, or literature information about the tissue. The
98 hyper-parameters controlling y_s can estimated based on total RNA counts in the input
99 data and the quality of the experiment (Section 1.2).

100 The prior distributions of $z_{s,r}$ and $x_{r,f}$ are defined as stated below:

- 101 • The total spatial abundance $z_{s,r}$ of latent cell type groups r across locations s is
102 defined as Gamma distributed with a prior controlling the scale and sparsity of $z_{s,r}$:

103

$$104 \quad z_{s,r} \sim \text{Gamma}(B_s/R, 1/(N_s/B_s)), \quad (6)$$

105 where N_s corresponds to the latent average number of cells in each location, and B_s
106 is the latent number of groups r expected in each location s (See eq. (7)-(8) below).
107 Unobserved B_s and N_s parameters are modelled as Gamma-distributed:

$$108 \quad N_s \sim \text{Gamma}(\hat{N} \cdot v^n, v^n) \quad (7)$$

109

$$110 \quad B_s \sim \text{Gamma}(\hat{B}, 1), \quad (8)$$

where $\hat{N}$ is a user-defined estimate of the expected number of cells per location; $\hat{B}$ is the expected average number of cellular compartments / zones per location (default=7); and v^n denotes the prior strength. Recommendation on setting $\hat{N}$ and v^n hyper-parameters are outlined in section 1.2 (keep $\hat{B}$ at default values).

This is constructed such that $\sum_r z_{s,r} \geq N_s$, i.e. the expectation of the sum over $z_{s,r}$ equals to the expected number of cells per location $\sum_r z_{s,r} = N_s$, and that on average each location has high value of $z_{s,r}$ for B_s expected cell type groups.

- The latent variable $x_{r,f}$ represents the contribution of each latent cell type group r to the abundance of each cell type f and is Gamma distributed with a prior controlling the number of cell types f that have high values of $x_{r,f}$ in each group r :

$$x_{r,f} \sim \text{Gamma}(K_r/R, K_r), \quad (9)$$

where K_r represents the unobserved number of cell types for each group r . This prior controls the absolute values of $x_{r,f}$ such that on average $\sum_r x_{r,f} = 1$.

K_r is Gamma-distributed with a prior informed by default hyperparameters:

$$K_r \sim \text{Gamma}(\hat{A}/\hat{B}, 1), \quad (10)$$

where $\hat{A}$ is expected average number of cell types per location (default=7); and $\hat{B}$ is expected average number of cellular compartments / zones per location (default=7, see section 1.2). This prior tells that each group r has large value of $x_{r,f}$ for many cell types f when $\hat{A} > \hat{B}$, and $\hat{A} = \hat{B}$ corresponds to the prior belief that the spatial abundance of each cell type f is independent from other cell types.

2. **Gene-specific multiplicative scaling factor m_g** is modelled as Gamma-distributed (Eq. (11) - (14)) with hierarchical prior μ^m and α^m which provide regularisation:

$$m_g \sim \text{Gamma}(\alpha^m, \alpha^m/\mu^m) \quad (11)$$

$$\alpha^m = 1/(o^m)^2 \quad (12)$$

$$o^m \sim \text{Exponential}(3) \quad (13)$$

$$\mu^m \sim \text{Gamma}(1, 1). \quad (14)$$

3. **The prior on detection efficiency y_s per location** is selected to discourage over-normalisation, such that unless data has evidence of strong within-experiment variability in RNA detection sensitivity across locations, it is assumed to be small and close in values to the mean sensitivity for each experiment/batch y_e (See Fig S5, Fig S8F):

$$y_s \sim \text{Gamma}(\alpha^y, \alpha^y/y_e), \quad (15)$$

where α^y is a user-provided hyper-parameter that regularises within-experiment variation (Section 1.2); and y_e is unknown/latent average detection efficiency in each batch/experiment:

$$y_e \sim \text{Gamma}(10, 10/\mu^y), \quad (16)$$

where μ^y is estimated using data input to the model ($d_{s,g}$, $g_{f,g}$) and the expected number of cells per location $\hat{N}$ (Section 1.2).

4. **Overdispersion for each gene.** Containment prior² is used for modelling unobserved variance using NB over-dispersion $\alpha_{e,g}$. The idea of a containment prior is that, by the prior, NB distribution should be maximally close to Poisson distribution, which is achieved by putting most probability mass on large $\alpha_{e,g}$. So, under the following prior most genes have low over-dispersion:

$$\alpha_{e,g} = 1/o_{e,g}^2 \quad (17)$$

$$o_{e,g} \sim \text{Exponential}(\beta^o) \quad (18)$$

$$\beta^o \sim \text{Gamma}(9, 3) \quad (19)$$

where constants 9 and 3 correspond to the prior $\alpha_{e,g}$ close to that observed in previous studies in scRNA-seq data³.

5. **Additive shift for genes.** $s_{e,g}$ represent background RNA counts for every gene g in every spatial experiment e and are modelled as Gamma distributed with hierarchical experiment-specific priors α_e^s and β_e^s which provide regularisation:

$$s_{e,g} \sim \text{Gamma}(\alpha_e^s, \alpha_e^s/\mu_e^s) \quad (20)$$

$$\mu_e^s \sim \text{Gamma}(1, 100) \quad (21)$$

$$\alpha_e^s = 1/o_e^2 \quad (22)$$

$$o_e \sim \text{Exponential}(\beta^s) \quad (23)$$

$$\beta^s \sim \text{Gamma}(9, 3) \quad (24)$$

where hierarchical prior α_e^s and μ_e^s model variation in background across experiments e ; and β^s serves as hyper-prior allowing additional flexibility for the model to learn the scale of α_e^s rather than relying on user input. These modelling choices follow the containment prior principle outlined above.

Joint modelling of multiple spatial experiments/batches provides the several benefits due to normalisation and sharing of information between experiments:

1. Modelling differences in RNA detection sensitivity across experiments - via hierarchical prior y_e on normalisation parameter y_s - allows direct comparison of cell abundance across multiple tissue sections with varying sequencing depth.
2. Increasing accuracy by improving the ability of the model to distinguish low sensitivity m_g from zero cell abundance $w_{r,f}$, which is achieved by sharing the variable representing gene-specific sensitivity m_g across experiments. Similarly to common practice in single cell data analysis, this is equivalent to regressing out the effect of technology but not the effect of individual experiment.
3. Increasing sensitivity by sharing factorisation prior on cell abundance $w_{r,f}$, namely which cell types tend to co-locate across experiments represented by $x_{r,f}$.

191 In addition to reporting absolute cell density we also compute absolute number of mRNA
 192 molecules ($u_{s,f}$) contributed by each cell type f to each location s :

$$193 \quad u_{s,f} = w_{s,f} \left(\sum_g m_g g_{f,g} \right) \quad (25)$$

194 $u_{s,f}$ are a more robust estimate than either absolute cell density $w_{s,f}$ or relative cell proportions
 195 $w_{s,f} / \sum_f w_{s,f}$. For this reason it can be used as a criteria to accurately exclude non-mapped cell
 196 types. For comparison of cell and mRNA abundance in the mouse brain see Suppl Files 1 and 6.
 197

198 1.2 Determining hyper parameter values

199 There are 3 parameters that are linked to the properties of the data. The remaining hyper-
 200 parameters should be used at their default values as specified in the next section.

201 The first two priors should be specified by the user for every tissue and the appropriate
 202 settings for the third parameter are derived by the software from input data.

203 1. **Expected cell abundance $\hat{N}$ per location** is a tissue-level global estimate, which can
 204 be derived from histology images (H&E or DAPI), paired to the spatial expression data
 205 $d_{s,g}$, or when not available - coming from the same tissue type. An estimate can be
 206 obtained by manually counting nuclei in a 10-20 locations in the histology image (e.g.
 207 using 10X Loupe browser, Fig S8A-B). When suitable histology images are not available,
 208 the size of capture regions relative to the size of cells can be used to estimate $\hat{N}$ (Slide-Seq
 209 V2, Fig S27).

210 For all analysis in this manuscript, a single tissue-level estimate was used.

211 2. **Hyperparameter α^y for regularising within-experiment variation in RNA detection**
 212 **sensitivity.** By default, we assume that the data does not have strong within-experiment
 213 variability in RNA detection sensitivity across locations ($\alpha^y = 200$), which results in values
 214 of y_s close to the mean sensitivity for each experiment y_e . However, when strong gradients
 215 in mRNA detection sensitivity are observed, a common issue in human adult 10X Visium
 216 data, we recommend less strict regularisation ($\alpha^y = 20$) to allow normalisation of such
 217 effect and improved sensitivity (data not shown).

218 3. **Expected detection sensitivity μ^y** is a single global estimate which is derived from
 219 average total RNA counts observed in input spatial data $d_{s,g}$, reference cell type signatures
 220 $g_{f,g}$, and the value of $\hat{N}$ as follows:

$$221 \quad \mu^y = \frac{\frac{\sum_s \sum_g d_{s,g} / S}{\hat{N}}}{\sum_f \sum_g g_{f,g} / F} \quad (26)$$

222 where S is the total number of locations and F is the total number of cell types.

223 While good choices of these hyper-parameters can have a positive impact on accuracy, the
 224 estimate of relative cell abundance and absolute mRNA abundance is robust to a range of
 225 reasonable hyper-parameters choices (Fig S5).

226 Slide-Seq V2 locations are expected to have fractions of cells in each location (“bead”),
 227 where we observed 1-2 cell types per location (Fig S12)⁴. However, we observed large absolute
 228 cell abundance $w_{s,f}$ of certain cell types in a fraction of location in the Slide-Seq V2 data (Fig

229 2H) potentially indicating that additional technical effects may need to be accounted for.

230
231 The following priors should be used at their default values unless stated otherwise:

- 232 1. The value of **the number of cell types** $\hat{A} = 7$ **per location** and **the number of**
233 **tissue zones** $\hat{B} = 7$ **per location**. The value of these hyper-priors has minimal effect on
234 model accuracy (Fig S5). Consequently, we recommend setting their values to 7, a single
235 global estimate, which is in the middle of hyper parameter space that was assessed in the
236 sensitivity analysis.
- 237 2. **The number of co-located cell type groups is** $R = 50$. Regardless of selected R the
238 model tends to find much lower number of groups (3-7) r that have substantial values of
239 $x_{r,f}$.
- 240 3. **Hyper-parameter** v^w , controlling the extent to which the co-abundance prior controls
241 $w_{s,f}$, is set to a medium value of $v^w = 5$. Increasing v^w forces stronger linear dependencies
242 in $w_{s,f}$ leading to over-smoothing and lower model accuracy. Decreasing v^w leads to cell
243 types f modelled as independent, which also decreases model accuracy because of the
244 extent of those effects in real data.
- 245 4. v^n **controlling the strength of $\hat{N}$ prior** in eq. (8) is set to 1 when a global rather than
246 location-specific estimate $\hat{N}$ is specified.

247 Optionally, as advanced feature, cell2location can utilise the per-location number of cells
248 that is determined by using image segmentation to estimate the number of nuclei (See Section
249 1.4, Fig S27). It should be provided as a location-specific $\hat{N}_s$ hyper-parameter and v^n should
250 be increased to make the prior more informative ($v^n = 10$). When the size of cell capture
251 areas varies substantially, such as in laser capture microscopy (LCM) method, we recommend
252 to provide a location-specific $\hat{N}_s$ estimate.

253 1.3 Inference

254 Variational Bayesian Inference is used to approximate the posterior, building on the Automatic
255 Differentiation Variational Inference (ADVI) framework implemented in pyro⁵. An alternative
256 implementation in pymc3⁶ is provided to simplify prototyping. Briefly, within the ADVI
257 inference framework, the posterior distribution over unknown parameters are approximated by
258 softplus-transformed (to ensure a positive scale) univariate normal distributions. The parameters
259 of these variational distributions are determined by minimizing the KL divergence between
260 the variational approximation and the true posterior distribution, or equivalently maximizing
261 the evidence lower bound (ELBO loss function). Training is stopped when the ELBO stops
262 increasing, usually after 20,000 - 50,000 iterations of training using ADAM optimiser (learning
263 rate 0.002). Posterior mean, standard deviation, 5% and 95% quantiles for each parameter are
264 computed using 1,000 samples from the Variational posterior distribution. We observed that
265 5% quantile provides a slightly more accurate estimate of cell abundance compared to the mean
266 so we used these estimates in all paper figures.

267 Using pymc3 implementation, we empirically observed that the estimated posterior distribution
268 are very stable to initial conditions (training restarts). Hence, we did not consider a model
269 selection step across restarts and did not implement this step in pyro. In principle training
270 restarts can be used as a quality control step. Significant divergence of the solutions between
271 restarts tends to indicate 1) poor match of the cell type reference or 2) large mismatch between
272 the prior distribution of $d_{s,g}$ (sample from the model given hyper-parameter choices) and the

count distribution in the observed data $d_{s,g}$ (prior predictive check). Optionally, the suitability of the selected hyper-parameters can be assessed using prior predictive check (also provided within the cell2location package in pymc3).

As a quality control step, we recommend evaluation of the model fit as described in [the tutorial](#):

1. comparing the posterior mean estimate $\log_{10}(\mu_{s,g} + 1)$ to observed data $\log_{10}(d_{s,g} + 1)$ (posterior predictive check);
2. assessing that y_s has successfully normalised technical RNA detection effect: 1) variation in estimated total cell abundance $\sum_f w_{s,f}$ should be consistent with histology (areas with more cells should have higher abundance) and 2) spatial distribution of y_s should generally resemble total RNA count $\sum_g d_{s,g}$ across locations.

1.4 Nuclei Segmentation (optional)

To perform nuclei segmentation, H&E-stained image of the Visium-profiled tissue was used. The image was acquired using Hamamatsu slide scanner as described in the section on experimental processing. Full resolution images were divided into sub-images of size about 1000 by 1000 pixels. Then a state-of-the-art convolutional neural network (CNN)-based segmentation pipeline described in Caicedo et al.⁷ was used. The model is available in GitHub repository [selimsef/dsb2018_topcoders](#) and the code to use it in [yozhikoff/segmentation](#). An ensemble of 32 pre-trained CNNs each with Unet- or FPN-like architecture was used to classify pixels to 3 classes: background, nuclei and nuclei boundaries. Predicted segmentation masks were averaged across CNNs, then a watershed algorithm was applied to refine predicted boundaries and separate individual nuclei. As a final step, a gradient boosting model over morphological features such as nuclei colour, shape and size was used to remove false-positively detected nuclei. Obtained individual nuclei masks were used to compute several morphological features: 1) area (count of pixels in a mask), 2) shape - ratio of lengths of the major and minor axis of an ellipse fitted to the mask, and 3) position - center of mass of the image containing individual mask. Kd-tree approach was used to efficiently assign nuclei to Visium locations. Kd-tree implementation from the scikit-learn package was used. Nuclei positions in the histology image are reported in the GitHub repository [vitkl/cell2location_paper](#).

2 Estimation of reference expression signatures of cell types

We provide two methods for deriving reference expression signatures of cell types $g_{f,g}$ from scRNA-seq expression data $J = \{j_{c,g}\}$ of each gene $g = \{1..G\}$ in cell $c = \{1..C\}$. For both methods, we assume that cell type labels for each cell are provided by the user, for example as obtained using conventional cell clustering to identify putative cell types and sub-populations.

First, untransformed and unnormalised count matrix $J_{c,g}$ is filtered to select expressed genes (Fig S1) at 2 cut-offs: 1) selecting genes detected at mRNA count > 0 in many cells ($> 5\%$ of cells), 2) selecting genes detected at mRNA count > 0 in a few cells ($5\% > \text{cell count} > 10$) but with mean expression across non-zero cells slightly larger than 1 (> 1.12). The cut-offs listed are a guideline, with values adjusted for every data set (Methods).

Next, we offer two alternative methods to derive $g_{f,g}$ from the scRNA-seq count matrix. Although method 1 is computationally more efficient, we recommend method 2 as robust general-purpose solution which provides more accurate results when the scRNA-seq data suffers from batch effects (Fig S24B):

1. **Model-based approach estimation of reference expression signatures of cell types $g_{f,g}$ using a regularised Negative Binomial regression.** This model robustly derives reference expression signatures of cell types $g_{f,g}$ using the data composed of multiple batches $e = \{1..E\}$ and technologies $t = \{1..T\}$. Adapting the assumptions of a range of computational methods for scRNA-seq^{8,9}, we assume that the expression count matrix $J = \{j_{c,g}\}$ follows a Negative Binomial distribution with unobserved expression levels (rates) $\mu_{c,g}$ and a gene-specific over-dispersion $\alpha_{e,g}$:

$$J_{c,g} \sim \text{NB}(\mu_{c,g}, 1/\alpha_g^2). \quad (27)$$

We model $\mu_{c,g}$ as a linear function of reference cell type signatures and technical effects:

$$\mu_{c,g} = (g_{f,g} + b_{e,g}) h_e p_{t,g}. \quad (28)$$

Here, h_e denotes a multiplicative global scaling parameter between experiments/batches e (e.g. differences in sequencing depth); $p_{t,g}$ accounts for multiplicative gene-specific difference in sensitivity between technologies which can be also used to regress out the effect of other covariates (such as donor effect); $b_{e,g}$ accounts for additive background shift of each gene in each experiment e (proxy for free-floating RNA).

The priors on these parameters are set as described below.

2. **Hard coded approach for computing mean count each gene g in cell cluster $f = \{1..F\}$.** This method is computationally highly efficient and generally yields good mapping quality as long as the scRNA-seq reference data are obtained from a single batch (e.g. 10X experiment) or several batches with minimal technical differences (Fig S24A). We also recommend this methods when using Smart-Seq 2 scRNA-seq data, as the distributional assumptions of method 2 are not appropriate for this technology.
3. **The software can utilise user-provided reference expression signatures $g_{f,g}$,** such as estimated with alternative methods and methods that could be developed in the future to account for more complex batch structures and cell type hierarchies (Discussion).

Details of the model-based estimation of reference expression signatures of cell types. Similarly to the cell2location model, the signature model is defined in a Bayesian framework, relying on ADVI inference and implemented in pyro. The following priors are used on parameters listed in eq. 28:

$$g_{f,g} \sim \text{Gamma}(1, 1), \quad (29)$$

$$h_e \sim \text{Gamma}(1, 1), \quad (30)$$

$$p_{t,g} \sim \text{Gamma}(200, 200/1). \quad (31)$$

where the constant 200 is chosen to provide regularisation and penalise deviation of $p_{t,g}$ values from 1, similarly to the approach used for y_s prior in eq. 15-16.

The prior on $b_{e,g}$ is specified identically to $s_{e,g}$ in eq. 20-24. The prior on α_g is specified similarly to $\alpha_{e,g}$ in eq. 17-19 as follows:

$$\alpha_g = 1/o_g^2 \quad (32)$$

$$g \sim \text{Exponential}(\beta^o) \quad (33)$$

$$\beta^o \sim \text{Gamma}(9, 3) \quad (34)$$

where constants 9 and 3 correspond to the prior $\alpha_{e,g}$ being close to that observed in previous studies in scRNA-seq data³.

$g_{f,g}$ is initialised at analytical average for each cell type f . Such informative initialisation leads to fast convergence. Implementation of training using mini batches makes the model fast (10min-60min on GPU) and scalable to very large data sets (> 200k cells). Training hyper-parameters are used as specified in Section 1.3 with recommended mini-batch size of 2500. The model training should stopped at the number of epochs needed for convergence, which should be determined for every data set by the user by examining the history of ELBO loss.

To verify that the model successfully accounted to non-biological sample effects h_e , $p_{t,g}$ and $b_{e,g}$, a corrected expression matrix $J_{c,g}^{corrected}$ is obtained by normalising $J_{c,g}$:

$$J_{c,g}^{corrected} = J_{c,g} / (h_e p_{t,g}) - b_{e,g} \quad (35)$$

$J_{c,g}^{corrected}$ matrix is subjected to standard Scanpy workflow¹⁰ to verify the mixing of the cells from different samples and technologies within matching cells types (Fig S19B).

3 Comparison of cell2location with alternative methods

Cell2location is related to alternative methods, which have recently been proposed or developed in parallel (Table 1). While cell2location and alternative methods are all based on regression, there are important differences that set cell2location apart from existing methods. These include in particular:

1. Cell2location models linear dependencies in abundance of cell types across locations by employing a factorized prior on the cell abundance variable $w_{s,f}$ (See Section 1.1). This additional level of hierarchy in cell2location is a major contributor to the model performance (Fig S25D-E, see below), and is not considered by existing methods.
2. Cell2location allows for joint modelling of multiple batches/experiments, thereby 1) sharing statistical strength for estimating key parameters, in particular those mentioned in point 1 and the gene-specific difference in sensitivity between technologies m_g , and 2) enabling direct comparison of cell abundance across batches with varying RNA detection efficiency by utilising normalisation using latent variable y_s .
3. Cell2location is implemented within a Bayesian framework, which allows for leveraging prior information on key experimental parameters in a rigorous manner (also see Section 1.2). The modular implementation in both pymc3 and pyro frameworks will also facilitate future extensions, for example to adapt cell2location to future spatial technologies (e.g. recently developed Nanostring WTA model extension provided on [GitHub](#)).

For a tabular comparison of specific features see Table 1, which considers the features mentioned above and other criteria.

In addition, we present a direct assessment how the individual modelling choices in cell2location (points 4-6 in Table 1) contribute to the model performance (Fig S4, Fig S24, Fig S25). Specifically, we evaluate the importance of 1) prior factorisation of cell abundance $w_{s,f}$; 2) gene-specific technology sensitivity scaling m_g (in both cases we observed substantial improvement

in accuracy); 3) NB regression for estimating reference signatures of cell types compared to hard-coded method (see Section 2).

This is assessed by simplifying the model presented in Section 1.1 in the following way:

1. **No $w_{s,f}$ factorisation model.** To estimate the impact of factorisation of cell abundance $w_{s,f}$ on accuracy (Fig S4, Fig S24, Fig S25E), we define a simplified cell2location model, where prior factorisation is replaced with the following hierarchical prior designed to enable meaningful comparison and as such allowing estimation of absolute density:

$$w_{s,f} \sim \text{Gamma}(A/F, 1/(N/A)), \quad (36)$$

where F corresponds to the number of cell types f , N corresponds to a single global estimate of the number of cells per location, A corresponds to a single global estimate of the number of cell types per location. The priors on N and A are defined as follows:

$$N \sim \text{Gamma}(\mu = \hat{N}, \sigma^2 = \hat{N}) \quad (37)$$

$$A \sim \text{Gamma}(\mu = \hat{A}, \sigma^2 = \hat{A}), \quad (38)$$

where $\hat{N}$ is a user-defined estimate of the expected number of cells per location, which is set to an appropriate value for a given tissue using the guidance in Section 1.2; $\hat{A}$ is the expected number of cell types (default $\hat{A} = 7$).

2. **No m_g model.** To estimate the impact of gene-specific technology sensitivity scaling m_g (Fig S4, Fig S24, Fig S25E) on accuracy we define a simplified cell2location model which does not use m_g but utilises all other parameters as described in Section 1.1.

3. **No NB regression approach.** We compare the two alternative strategies to estimate reference signatures (see Section 2, Fig S24, Fig S25E) in the context of the quantitative benchmarking analysis in the lymph node and the gut (See Methods), which demonstrates that the regression-based model results in higher mapping accuracy than the simpler hard-coded method (Fig S24), by combining data across technologies and batches (Fig S19).

We observed that the hard-coded method based on mean counts yields similar results to the model-based approach in the mouse brain snRNA-seq data, where the batch effects are relatively minor (Fig S24A). In contrast, the scRNA-seq data from human lymphoid organs has more substantial batch effects, as these data span 3 published studies from 3 organs, 21 experiments/batches, as well as 3' and 5' technologies (Fig S19). Consistently, we observed mapping of T CD4+ Tfh GC cells, but not T CD4+ Tfh and T CD4+ cells, to GC zone with model-based approach whereas hard-coded approach led substantial abundance of T CD4+ Tfh and T CD4+ cells in GC dark zone where those cell types should not be located according to the prior knowledge (Fig S24C, Methods).

These results indicate that the model-based approach is in general more accurate, in particular when the scRNA reference is composed of multiple batches and technologies.

4 Downstream analyses

The output of cell2location can be used as a starting point for a range of different downstream analyses. Here, we outline strategies how cell2location can be combined with conventional clustering, as well as non-negative matrix decomposition.

Table 1: Comparison of cell2location with other alternative methods

Criteria	cell2location	stereoscope ¹¹	Seurat V3 ¹²	SpotLIGHT ¹³	NNLS (autogenes) ¹⁴	RCTD ⁴
1. Data distribution, normalisation	Negative Binomial, non-negative parameters	Negative Binomial, non-negative parameters	Normalised $\log(x+1)$	Counts normalised by standard deviation, non-negative parameters	Counts, negative parameters	non-Negative Binomial, log-normal model
2. Inference method	Variational Inference	Penalised maximum likelihood	PCA + MNN	NNLS	NNLS	Penalised maximum likelihood
3. Method for estimating signatures	2 methods provided but user-derived signatures can be used (See Section 2)	Hard-coded method (NB regression with unusual parametrisation)	Not needed	Hard-coded method (topic model)	User-derived signatures	Hard-coded method (average gene expression)
4. Accounting for the difference in sensitivity between single cell and spatial technology	Yes	Yes	Unclear	No	No	Yes
5. Models similarity of locations between cell types (prior factorisation of $w_{s,f}$)	Yes	No	No	No	No	No
6. Method for estimating signatures accounts for technology and batch effects in scRNA data	Yes	No	Possible	No	No	No
7. Joint modelling of multiple spatial experiments	Yes	No	Unclear	No	No	No
8. Normalisation of RNA detection sensitivity	Yes, regularised using hierarchical priors	Yes, non-regularised	Unclear	No	No	Yes, non-regularised
9. GPU acceleration	Yes	Yes	No	No	No	No
10. Mini-batch training	Yes, at lower accuracy	Yes, likely at lower accuracy	No	No	No	No
11. Language	Python	Python	R	R	Autogenes in Python, other in R	R

4.1 Identification of tissue regions by clustering

The cell abundance $w_{s,f}$ estimated by cell2location model is used as input to cluster locations s into clusters representing non-overlapping regions. This can be achieved by constructing a KNN graph that represents connectivity of locations in terms of their cell composition (function `scanpy.pp.neighbors`). Reference implementations of Leiden clustering, such as the implementation (function `scanpy.tl.leiden`) in the Scanpy package¹⁰, can utilise cell composition KNN graph to group organisation of the tissue. In the experiments presented in the paper, Leiden clustering was performed with default settings except for the resolution parameter (see main text Methods). The resulting clusters were cross-referenced with the mouse brain anatomy using corresponding histology images and similar clusters were merged to generate the broad region map.

4.2 Groups of co-located cell types

As an alternative to clustering, we considered non-negative matrix factorisation (NMF), thereby identifying cellular compartments within the tissue. Briefly, in NMF, absolute cell abundances $w_{s,f}$ of each cell type f across locations s are modelled as an additive function of the cell type groups r (cellular compartments). The factorization assumption of NMF combined with the

additivity constraint allow for identifying cellular compartments, which unlike discrete clustering can co-occupy one common location (Fig S20). Briefly, cell abundance estimates $w_{s,f}$ are modelled as an additive decomposition:

$$w_{s,f} = \sum_r z_{s,r} x_{r,f} \quad (39)$$

where $x_{r,f}$ represents NMF weights describing the contribution of each cell type group r to cell types f ; $z_{s,r}$ represents NMF weights describing the abundances of each cell type group r across locations s .

Similarly to cell2location model (eq. (25)), we compute $u_{s,r}$ which can be intuitively interpreted as the number of cells from each cell type group r in each location s and use it for visualisation:

$$u_{s,r} = z_{s,r} \sum_r x_{r,f} \quad (40)$$

Inference is performed using standard NMF in the scikit-learn package¹⁵. The model is trained 2 times to evaluate stability of the identified cellular compartments. The first training iteration is always selected.

Rationale for applying the NMF to multi-cell spatial data. We hypothesise that linear dependencies in the abundance of cell types across locations reflects the interaction between these cell types, specifically, any interaction that could affect mutual location of these cell types. For this reason we tested this approach on human lymph nodes, a tissue where locations of cell types are determined by their interactions (Fig 4D). 10X Visium technology combines cell types located within the $55\mu m$ diameter of capture areas, a distance at which the action of local paracrine signals and direct adhesive contacts can be observed, likely driving linear dependencies in cell abundance. We assume that the signal does not propagate between neighbouring locations spaced by $45\mu m$ gaps, therefore using proximity information is not essential. Therefore we note that this interpretation of NMF decomposition of cell abundances is a unique feature of grid-based technologies that aggregate multiple cells and cell types within single locations. This interpretation does not directly apply to technologies with increased resolution (Slide-Seq V2) or technologies where capture regions represent selected features.

Potential cell interactions driving linear dependencies in cell type abundance can be classified into:

- Signals promoting proliferation and survival of other cell type
- Adhesion molecules that increase the frequency of contacts
- Recruitment signals such as CXCR5 / CXCL13¹⁶
- Repulsive signals, inhibiting the action of chemo-taxis signals such as PD-1 / PD-L1 interactions¹⁷
- Interactions that induce new transcriptional states

5 Constructing a synthetic spatial transcriptomics data set

This section outlines the mathematical description and the step-by-step procedure for constructing a synthetic spatial transcriptomics data set. The procedure is comprised of three steps: 1) simulating abundance of cell types across locations, 2) simulating expected multi-cell mRNA expression of genes across locations, 3) simulating multi-cell mRNA integer counts weighted by technology difference effect.

491 **First**, follow this step-by-step procedure to generate 1) ground truth spatial abundance
 492 $w_{s,f}$, 2) integer cell count $count_{s,f}$ and 3) fraction of mRNA captured $frac_{s,f}$ for every cell type
 493 f across locations s :

- 494 1. Assign cell types to ubiquitous (n=8) and regional (n=41) abundance patterns (denoted
 495 as r).
- 496 2. Perform binary assignment of 41 sparse cell types to 12 tissue zones and 8 ubiquitous cell
 497 types to 8 ubiquitous patterns (total n=20), shown in Fig S2A and denoted as $x_{r,f}$.
- 498 3. Stratified by location pattern, randomly assign up to 20% of cell types to high abundance
 499 groups and all other cell types to low abundance groups. Generate per cell type average
 500 abundance d_f , which is different for 4 groups shown in Fig 1B:

- (a) Ubiquitous and low abundance: 5 cell types present in most locations at abundance:

$$501 \quad d_f \sim \text{Gamma}(1.0 \cdot 5.0, 5.0). \quad (41)$$

- (b) Ubiquitous and high abundance: 3 cell types present in most locations at abundance:

$$504 \quad d_f \sim \text{Gamma}(2.8 \cdot 5.0, 5.0). \quad (42)$$

- (c) Sparse and low abundance: 32 cell types present in sparse tissue zones at abundance:

$$507 \quad d_f \sim \text{Gamma}(1.0 \cdot 5.0, 5.0). \quad (43)$$

- (d) Sparse and high abundance: 9 cell types present in sparse tissue zones at abundance:

$$510 \quad d_f \sim \text{Gamma}(2.8 \cdot 5.0, 5.0). \quad (44)$$

513 By following this procedure, sparsity and abundance parameters for each cell type were
 514 generated that produced an average total number of cells per location close to 10, mimicking
 515 cell count observed by nuclear segmentation of the mouse brain histology images (Fig S8,
 516 Suppl Methods). Per cell type maximum abundance d_f was used to scale $x_{r,f} = x_{r,f} * d_f$,
 517 thus defining the average abundance of each cell type across patterns r .

- 518 4. Next, generate spatial abundance $z_{s,r}$ for locations s for 20 location patterns (denoted as
 519 r) representing 12 tissue zones and 8 ubiquitous cell types. Gaussian Process in 50x50
 520 grid of locations was used to generate spatial abundance values with randomly generated
 521 bandwidth parameters:

$$522 \quad bw_r \sim \text{Gamma}(8.0 \cdot 1.2, 1.2). \quad (45)$$

523 Bandwidth controls how quickly cell abundance changes in spatial coordinates. Eta
 524 parameter controls the sparsity with large values of eta corresponding to high variance in
 525 Gaussian Process, putting distribution weight away from 0 and correspondingly producing
 526 sparse cell abundance maps.

- (a) 8 ubiquitous patterns with for non-zero abundance in most locations:

$$528 \quad z_{s,r} \sim \text{GP}(\mu = 0, eta = 0.5, bw = bw_r). \quad (46)$$

- (b) 12 tissue zones with sparse locations:

$$530 \quad z_{s,r} \sim \text{GP}(\mu = 0, eta = 1.5, bw = bw_r). \quad (47)$$

531 5. To ensure positive scale, cell abundance for each pattern $z_{s,r}$ were softmax-transformed:

$$532 \quad z_{s,r} = e^{z_{s,r}} / \sum_r e^{z_{s,r}}. \quad (48)$$

533 Next, to ensure that maximum for each location pattern r is equal to 1 further normalisation
534 was applied:

$$535 \quad z_{s,r} = z_{s,r} / \max_s z_{s,r}, \quad (49)$$

536 which is needed to use abundances established in step 3 as average value for each cell
537 type.

538 6. Per cell type abundance for each location s was generated as a weighted sum of contributions
539 of tissue zones (shown in Fig S2B):

$$540 \quad w_{s,f} = \left(\sum_r z_{s,r} x_{r,f} \right) q_{s,f}, \quad (50)$$

541 where $q_{s,f}$ introduces randomness to abundances of individual cell types:

$$542 \quad \log(q_{s,f}) \sim \text{Normal}(\mu = 0, \sigma = 0.35). \quad (51)$$

543 This additional variability mimics the observation that co-located cell types in real tissues
544 do not have perfectly correlated abundance within tissue zones.

545 7. Cell abundance $w_{s,f}$ was used to generate integer cell count $count_{s,f}$ and fraction of mRNA
546 captured $frac_{s,f}$ for each location and cell type as follows:

547 (a) generate integer cell count by rounding $\lceil w_{s,f} \rceil$ to the smallest integer, such that
548 $count_{s,f} \geq w_{s,f}$.

549 (b) Compute the fraction of mRNA captured as $frac_{s,f} = w_{s,f} / count_{s,f}$.

550 **Second**, follow this step-by-step procedure to generate expected multi-cell mRNA count
551 profiles $ed_{s,g}$ for every gene g in a given location s using 1) the integer cell count $count_{s,f}$ and
552 2) the fraction of mRNA captured $frac_{s,f}$ for each cell type f in a given location s (from the
553 steps above), by combining cells c drawn from reference cell types f in the snRNA-seq data $j_{c,g}$
554 as follows:

555 1. Create a matrix $a_{s,c}$ which that defines which cells should be added to construct every
556 location. Randomly select indices of cells c of a given cell type f ($c \in f$) to form a subset
557 s for every location s containing $n = count_{s,f}$ cells:

$$558 \quad count_{s,f} = \sum_s a_{s,c \in f}. \quad (52)$$

559 2. Construct per cell type expected mRNA abundance profiles for a given location and cell
560 type:

$$561 \quad ed_{s,f,g} = \left(\sum_c a_{s,c \in f} \cdot j_{c,g} \right) \cdot frac_{s,f}. \quad (53)$$

562 3. Construct multi-cell expected mRNA abundance profiles by adding mRNA across all cell
563 types:

$$564 \quad ed_{s,g} = \sum_f ed_{s,f,g}. \quad (54)$$

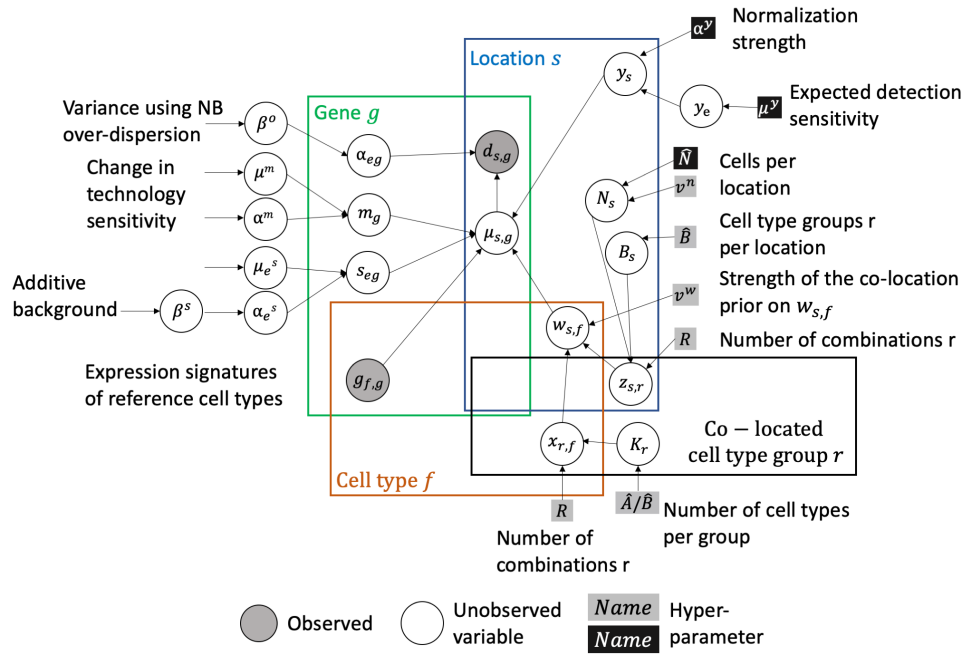


Figure 2: Illustration of the corresponding graphical model. Square boxes denote hyper-parameters, with black box highlighting the parameters that require adaptation to a specific data set (determined by tissue type and spatial technology), as mentioned in Section 1.2.

Third, follow this step-by-step procedure to generate multi-cell mRNA integer counts $d_{s,g}$. In this step, gene-specific scaling was applied (m_g) to mimic the difference in sensitivity between technologies and counts were samples from Poisson distribution:

$$d_{s,g} \sim \text{Poisson}(ed_{s,g} m_g), \quad (55)$$

where m_g characterises the difference between the mouse brain Visium data and single nucleus RNA-seq reference (Fig S2C, estimated by cell2location). Using these values makes the simulated data representative for mapping single nucleus RNA-seq derived reference cell types. Under this simulation, the total number of mRNA per location mimics that observed in mouse brain data (Fig S2D).

6 Appendix

6.1 Complete graphical model diagram

The complete illustration of the cell2location graphical model, complementing Figure 1, is shown in Figure 2.

6.2 Notation

Two alternative parameterizations of Gamma distribution are used: 1) α (also called shape or over-dispersion) and β (also called rate); 2) μ and σ^2 ; with the link between the two given by:

$$\mu = \alpha/\beta; \sigma^2 = \alpha/\beta^2 \quad (56)$$

582

583

$$\alpha = \mu^2/\sigma^2; \beta = \mu/\sigma^2 \quad (57)$$

584 By default, α and β parametrization is used; when μ and σ^2 are used their notation is shown.

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